

CampusHive: A Peer-to-Peer Collaboration and Knowledge-Sharing Platform for Academic Communities

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Abstract—Contemporary campus communication relies on fragmented, general-purpose platforms such as WhatsApp, Telegram, and institutional email, leading to information loss, unstructured academic discourse, and disconnected event workflows. This paper presents CampusHive, a unified, mobile-first peer-to-peer collaboration and knowledge-sharing platform engineered to consolidate academic doubt resolution, project-based team formation, and campus event management within a single application. Built on a Flutter frontend with a serverless Firebase backend, the system implements Firebase Authentication for secure multi-role access and Cloud Firestore for real-time data synchronization across Students, Forum Admins, and Super Admins. A like/dislike answer-quality mechanism surfaces high-utility responses within the Doubt Hub module, while a conditional skip-logic event registration pipeline prevents workflow overhead under concurrent load. Evaluation of the deployed prototype demonstrates measurably improved academic collaboration efficiency and a reduction in platform-switching overhead among student users.

Index Terms—Flutter, Firebase, Peer-to-Peer Learning, Event Management Systems, Real-Time Synchronization, Role-Based Access Control, Academic Collaboration

I. INTRODUCTION

Educational institutions consistently grapple with the challenge of coordinating academic interaction across dispersed, heterogeneous communication channels. Students typically resort to large WhatsApp or Telegram groups for doubt resolution, where valuable discussions are overwhelmed by off-topic noise and pertinent information is rapidly buried within chat histories. Simultaneously, project collaboration remains ad-hoc, lacking structured spaces for team formation and milestone tracking, while campus event announcements are distributed through fragmented email threads or physical notice boards disconnected from registration workflows.

Existing commercial platforms such as Stack Overflow, Slack, or Eventbrite address isolated aspects of these challenges but are designed for general audiences and fail to accommodate the multi-tiered administrative hierarchies and co-curricular governance requirements inherent to university environments [1], [2]. The resultant fragmentation imposes significant cognitive overhead on students and administrators alike, inhibiting effective knowledge exchange and collaborative learning outcomes.

To address this structural deficiency, this paper introduces CampusHive, a centralized, mobile-first student interaction ecosystem. The platform converges four discrete operational domains: (1) structured academic doubt discussion with community-driven answer quality signals, (2) project group formation with integrated team chat, (3) campus event discovery and registration, and (4) multi-role administrative governance. Built on Flutter and Firebase, CampusHive demonstrates how domain-specific engineering of real-time synchronization and role-based access control can deliver a robust, paperless collaborative learning environment.

II. RELATED WORK

A growing body of research highlights the effectiveness of peer-driven academic systems while simultaneously revealing gaps that motivate the present work. Table I summarizes key prior contributions reviewed during system design.

The collective gap across these works is the absence of a single, college-specific platform that unifies doubt clearing, project collaboration, event management, and structured knowledge sharing. Existing systems either address isolated concerns or remain theoretical without implementation artifacts suitable for direct institutional deployment. CampusHive

TABLE I
SUMMARY OF RELATED WORKS

Paper	Year	Methodology	Result	Limitation
Peer-Mentoring for Engineering Students	2024	Structured peer-mentoring; surveys & performance analysis	Improved performance, confidence, peer bonding	Needs continuous senior participation
Q&A for Collaborative Learning with Quality Predictor	2013	Web-based Q&A with heuristic answer scoring	Structured knowledge base; improved learning	No project collab or real-time interaction
Social Learning Networks	2014	Survey of social platforms with peer validation	Peer feedback & reputation systems raise quality	Theoretical; no college-specific implementation
Computer-Supported Collaborative Learning	2018	Digital platforms with group tasks & peer evaluation	Collaborative platforms improve problem-solving	Requires moderation; misinformation risk

is designed to close this gap through an integrated, mobile-native architecture.

III. SYSTEM ARCHITECTURE

CampusHive follows a decoupled, cloud-native, three-tier architectural paradigm engineered to minimize client-side computational load, enforce strict multi-role data isolation, and support horizontal scaling as user populations grow. The architecture spans a Flutter-based presentation layer, a Firebase serverless backend, and a Cloud Firestore NoSQL data infrastructure tier.

A. Presentation Layer (Flutter Frontend)

The client interface is built as a Flutter Single-Page Application targeting Android devices. Flutter’s reactive widget tree enables component-scoped state management, ensuring that administrative panels, student views, and forum admin dashboards are logically isolated through role-guarded navigation guards. The widget reconciliation engine computes minimal diff operations against the widget tree, preventing layout reflow bottlenecks during high-frequency real-time data streams from Firestore listeners.

B. Backend Layer (Firebase Services)

The backend is composed entirely of serverless Firebase services, eliminating operational server management overhead. Firebase Authentication handles user identity lifecycle management, supporting Email/Password and Google OAuth sign-in flows. Custom user role metadata (*student*, *forum admin*, *super admin*) is persisted as Firestore document fields, with Firestore Security Rules enforcing corresponding data access boundaries natively at the database layer. Firebase Storage manages binary assets for profile images and event media.

C. Data Infrastructure Tier (Cloud Firestore)

Structured application state is handled via a denormalized, NoSQL Firestore document-collection model. The schema deliberately avoids deeply nested relational structures to guarantee ultra-low-latency reads during traffic spikes. Four primary collection hierarchies are maintained:

- **Users Collection:** Root-level documents keyed by Firebase UIDs, storing profile attributes, college affiliations, department metadata, and indexed arrays referencing joined project groups and registered events.
- **Events Collection:** Parent documents managed by Forum Admins, containing state vectors for title, category, capacity limits, registration deadlines, and fee parameters.
- **Doubts Collection:** Parent documents for academic questions with nested Replies sub-collections containing answer text, author references, and aggregated vote counters.
- **Projects Collection:** Group documents encapsulating metadata, technology stack tags, member capacity, and embedded real-time chat message sub-collections.

IV. MODULE DESCRIPTION

CampusHive is partitioned into four functionally cohesive modules, each addressing a distinct academic operational domain. The module boundaries mirror the top-level navigation structure exposed to end users and the administrative actor hierarchy.

A. User Management Module

This module governs the complete identity lifecycle: account registration, multi-role authentication, profile configuration, and session management. Upon initial registration, users configure a structured profile comprising their college affiliation, department, roll number, and a multi-select field-of-interest taxonomy spanning Flutter, React, ML/AI, Firebase, and Blockchain. The role assignment pipeline distinguishes three actor classes: Community Members (students), Forum Admins (club leads), and Super Admins. Forum Admin accounts require a Super Admin approval workflow before privileges are activated, preventing unauthorized escalation.

B. Doubt Discussion Module (AskHub)

The Doubt Hub provides a structured Q&A environment modeled on community-driven knowledge platforms. Students post tagged academic doubts with markdown-supported question bodies. Peers and seniors submit threaded answers immediately visible to all authenticated users. A symmetric like/dislike rating mechanism aggregates community signals on answer utility, surfacing high-quality responses through ranked presentation. Questions are classified into Hot, New, and Unsolved feed tabs. A trending topics panel surfaces hashtag-indexed subject clusters by active doubt count, and a top questions leaderboard exposes the most community-validated discussions.

C. Project Collaboration Module

This module enables structured team formation aligned with real campus project initiatives. Project owners publish group cards specifying the project title, technology stack, maximum member capacity, and cover image. Interested students submit join requests, which the project owner approves or rejects through an integrated management panel. Upon admission, members gain access to a persistent group chat portal supporting text messages and image attachments. Ownership transfer functionality allows project creators to delegate administrative rights to other members, ensuring continuity when original owners disengage.

D. Event Management Module

Forum Admins publish campus events to a categorized discovery feed spanning Technical, Cultural, Sports, and Workshop classifications. Each event card exposes structured metadata: date, time, venue, available seat count, registration fee, delivery mode, and a registration deadline countdown. Students interact with the event detail view to register and, for paid events, upload payment confirmation documents through a conditional skip-logic workflow. Super Admins retain full event oversight through a cross-forum administrative dashboard.

V. METHODOLOGY & TECHNICAL IMPLEMENTATION

A. Real-Time State Synchronization

To maintain presentation layer consistency across concurrent active sessions without performance-heavy HTTP polling, CampusHive’s synchronization architecture leverages Firestore’s Observer pattern implementation. Flutter widgets register persistent Firestore snapshot listeners that establish long-lived data channels to the document repository. When any cloud document is mutated by a concurrent writer, the database engine computes a micro-delta containing only modified fields and pushes the update through the active listener, triggering a targeted widget re-render without requiring full page reloads. This architecture is critical for the group chat portal and event seat-count displays.

B. Role-Based Access Control via Firestore Security Rules

Data access governance is enforced declaratively through Firestore Security Rules evaluated server-side before any read or write operation is committed. Role attributes persisted in the Users collection document are referenced within rule expressions to authorize or reject requests based on the authenticated user’s role vector. This design ensures that authorization checks execute at $O(1)$ complexity relative to user count and remain structurally enforceable even if client-side application code is bypassed. Administrative operations such as event creation and user management are gated behind explicit role checks that cannot be circumvented from the client layer.

C. Conditional Skip-Logic Registration Pipeline

To handle workflow variations between paid and free campus events, the registration engine implements a deterministic boolean routing function. When a user initiates registration for event instance x , the application evaluates:

$$P(x) = \begin{cases} 1 & \text{if } x.isPaid = \text{false} \\ 0 & \text{if } x.isPaid = \text{true} \end{cases} \quad (1)$$

If $P(x) = 1$, the pipeline clears payment verification overhead and immediately appends the user record to the confirmed attendee sub-collection. If $P(x) = 0$, a state-machine interceptor suspends the registration context and transitions the view to a payment document upload screen, preventing further state progression until binary transaction metadata is committed to Firebase Storage.

D. Answer Quality Ranking via Community Voting

The Doubt Hub implements a symmetric voting mechanism where each authenticated user may cast a single upvote or downvote against any answer. Vote states are persisted as keyed boolean maps within the answer document, preventing duplicate voting without requiring server-side transaction locks. The net vote score (upvotes minus downvotes) serves as the primary sort key when rendering answers in the default view, ensuring that community-validated responses are surfaced preferentially over chronologically recent but lower-utility submissions.

VI. SOFTWARE REQUIREMENTS SPECIFICATION

Table II consolidates both functional (FR) and non-functional (NFR) requirements formally identified during the requirements analysis phase.

VII. RESULTS & EVALUATION

The CampusHive prototype was deployed and evaluated within a controlled academic environment. The implemented system spans four navigational domains rendered through a bottom navigation bar: Projects, Doubt Hub, Events, and Profile.

A. Authentication and Profile Configuration

The login interface supports both email/password credentials and Google sign-in continuity. First-time users are routed to a structured profile setup wizard that collects college affiliation, department, batch year, roll number, and a multi-select technology interest taxonomy. The onboarding flow demonstrated zero field-skip validation failures across tested sessions, ensuring profile completeness prior to platform access.

B. AskHub Doubt Discussion Feed

The Doubt Hub rendered a live feed of questions categorized into Hot, New, and Unsolved tabs with sub-second update latency upon new question publication. The trending topics panel accurately reflected real-time hashtag distribution, and the top questions leaderboard correctly ranked entries by aggregate vote score. Answer submission and vote state

TABLE II
SOFTWARE REQUIREMENTS SPECIFICATION

ID	Requirement	Description
FR-1	User Authentication	Students register and log in securely via email/password or Google OAuth through Firebase Authentication.
FR-2	Doubt Posting	Users post academic doubts tagged by subject area with markdown-supported content.
FR-3	Doubt Answering	Students and seniors submit threaded answers to posted doubts.
FR-4	Answer Rating	Users upvote or downvote answers to surface high-quality responses.
FR-5	Project Group Management	Students create project groups with member capacity and technology stack definitions.
FR-6	Join Request Workflow	Users request to join groups; the project owner approves or rejects requests.
FR-7	Group Chat	Confirmed project members communicate via built-in real-time group chat.
FR-8	Event Discovery	Students browse events categorized as Technical, Cultural, Sports, or Workshop.
FR-9	Event Registration & Payment	Students register for events and submit payment proof through the platform.
FR-10	Admin Dashboard	Super Admins approve Forum Admin accounts, manage content, and view analytics.
NFR-1	Security	All transmissions use HTTPS; Firestore Security Rules enforce role-based data isolation.
NFR-2	Usability	The mobile app complies with Material Design guidelines for intuitive navigation.
NFR-3	Performance	UI interactions respond within 300ms under normal network conditions.
NFR-4	Scalability	Firebase/Firestore backend supports horizontal scaling for growing user loads.
NFR-5	Reliability	System targets 99.5% uptime leveraging Firebase’s managed infrastructure SLA.

persistence were validated across concurrent sessions with consistent state convergence.

C. Project Collaboration and Group Chat

The project discovery feed successfully filtered listings by technology category. Join request workflows demonstrated correct state transitions from Pending through Approved across both the owner’s management view and the applicant’s profile panel. The group chat portal delivered message persistence and real-time delivery with observable sub-second latency on standard 4G network conditions. Ownership transfer operations executed atomically with correct role reclassification.

D. Event Management and Registration

Events published by Forum Admins appeared immediately on the student discovery feed with correctly computed seat availability counters. The conditional registration pipeline

routed free events to immediate confirmation and paid events to the payment document upload screen without exception. The Super Admin dashboard correctly surfaced pending Forum Admin account requests with approve/reject workflow controls.

VIII. SOCIETAL ALIGNMENT & SUSTAINABILITY

The development objectives of CampusHive directly align with three United Nations Sustainable Development Goals:

- **SDG 4 (Quality Education):** By centralizing doubt resolution and enabling peer mentoring at scale, the platform widens access to on-demand academic support beyond formal classroom hours, reinforcing inclusive and equitable quality education within institutional communities.
- **SDG 9 (Industry, Innovation, & Infrastructure):** The serverless, cloud-native architecture modernizes campus administrative infrastructure, replacing paper-dependent event workflows with resilient digital pipelines capable of handling institutional-scale concurrent user loads.
- **SDG 17 (Partnerships for the Goals):** The project collaboration module structurally incentivizes interdisciplinary team formation, fostering cooperative partnerships necessary for knowledge-economy skill development in engineering education.

IX. CONCLUSION AND FUTURE SCOPE

CampusHive successfully implements a resilient, mobile-first paradigm for campus academic collaboration and institutional event administration. By converging structured doubt resolution, project team formation, integrated group communication, and event lifecycle management within a unified Flutter/Firebase application, the platform demonstrates a substantial improvement over fragmented, general-purpose communication channels. The community-driven answer quality mechanism, real-time state synchronization engine, and role-gated administrative workflows collectively validate the feasibility of domain-specific engineering for academic digital ecosystems.

Future development iterations will focus on the following enhancements:

- **AI Chatbot Integration:** An LLM-powered doubt answering agent to provide instant responses for frequently asked academic questions.
- **Video Conferencing:** Embedded WebRTC-based video meeting functionality for distributed project team synchronization sessions.
- **Real-Time Push Notifications:** Firebase Cloud Messaging integration for event deadline reminders, doubt reply alerts, and join request status updates.
- **Recommendation Engine:** A collaborative filtering model suggesting relevant project groups and open questions based on declared interests and interaction history.
- **End-to-End Encrypted Private Messaging:** User-to-user private chat with cryptographic message confidentiality guarantees.

- **Event Feedback & Report Vault:** Post-event feedback workflows and accreditation document archival for institutional audit compliance.

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